9 **D** frictional forces on car tires provide centripetal force

$$f_{max} = \frac{mv^2}{r}$$

$$f'_{max} = \frac{mv_{wet}^{\ 2}}{r}$$

$$\frac{f_{max}}{2} = \frac{mv_{wet}^{\ 2}}{r}$$

$$\frac{mv^2}{2r} = \frac{mv_{wet}^{\ 2}}{r}$$

$$v_{wet} = \frac{v}{\sqrt{2}} = \frac{30}{\sqrt{2}} = 21.213 = 21\ \text{m s}^{-1}$$